Road Case Study: Geosynthetic-Encased Columns

GEO-31 Engineering Geology | Short study handout

Main Idea

The document explains how geosynthetic-encased columns (GECs) help road embankments built over soft soil. A GEC is a sand or granular column wrapped with geosynthetic material. The wrap gives lateral support, so the column can carry load without spreading too much inside weak clay.

Why It Matters For Roads

Soft clay under a road can create excessive settlement, lateral deformation, and instability. GECs improve this situation because they share the embankment load, reduce settlements, and create faster drainage paths for excess pore water pressure.

Case Context

- Real case connected to road work near Sao Jose dos Campos, Sao Paulo.
- The ground had soft clay layers, which are difficult for road embankment construction.
- The solution used sand or granular columns encased by geosynthetic sleeves.
- Installation displaced the soil, generated pore pressure, and then allowed consolidation with time.

Technical Point

The study separates consolidation into two moments: column installation and embankment construction. This matters because settlement is not only a final value. It changes with time as pore pressure dissipates.

Key lesson

In road geotechnics, the solution must control both stability and time-dependent settlement. GECs are useful because they combine reinforcement, drainage, and load transfer.

Use In Your Work

This case can explain why soft soil treatment is necessary before road construction. It also shows how a geotechnical solution is chosen from soil behavior: weak clay needs reinforcement, drainage, and settlement control.

Source: Rational analysis of consolidation in geosynthetic encased columns foundation system.pdf